CHAPTER 1-3

Example 1.15

① id is a sequence of letters, digits, underscores

② id must start with a letter or underscore

③ id allow upper & lower case letters.

<id> → <letter> <rest> / <undscr><rest>

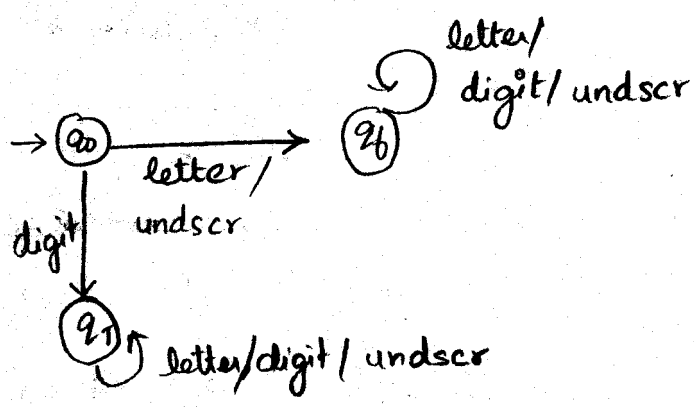
<rest> → <letter><rest> / <digit><rest> / <undscr><rest> / λ

<letter> → a|b|c| - - - z | A|B|C - - Z

<digit> → 0|1|2 - - - |9

<undscr> → _

1.16



Example 1.17

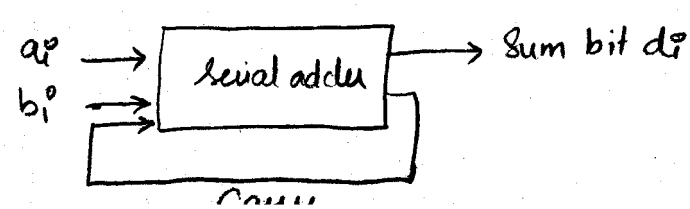
Binary Adder

a_i \ b_i	0	1
0	0 No Carry	1 No Carry
1	1 No Carry	0 Carry

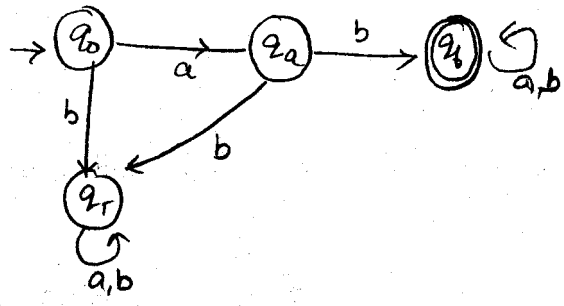
$$v(x) = \sum_{i=0}^{n} a_i 2^i$$

i/p : (a_i, b_i)

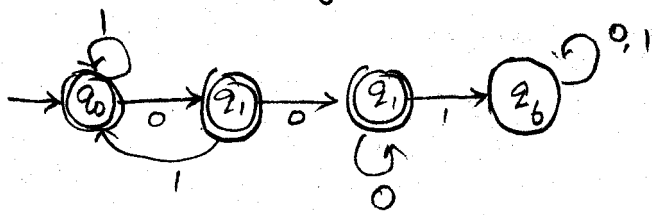
o/p = sum bit d_i



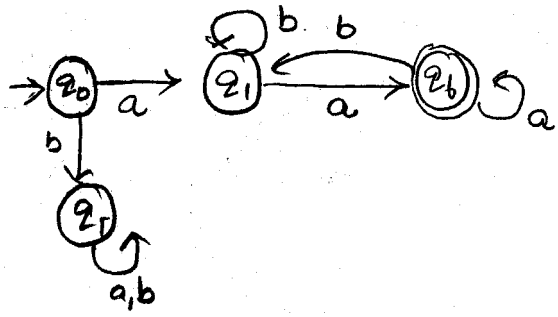
Example 2.3 Find Dfa that recognises all strings on $\Sigma = \{a,b\}$ with prefix ab.



Example 2.4 Find a dfa that accepts all the strings on $\{0,1\}$ except those containing the substring 001.



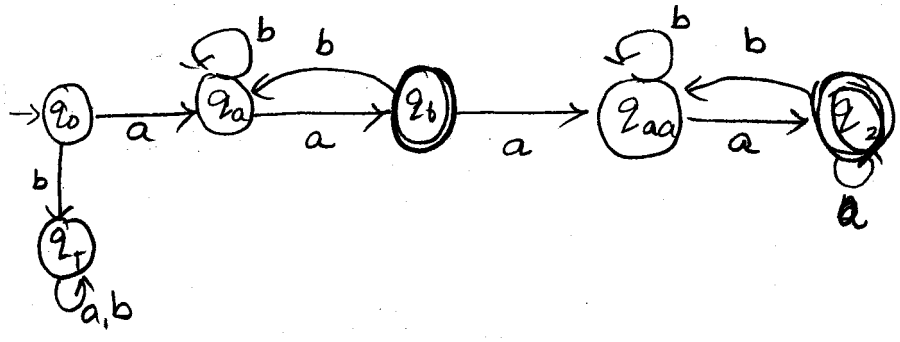
Example 2.5 Show that $L = \{awa : w \in \{a,b\}^*\}$ is regular.



Show that Regular $\Rightarrow$ Draw Dfa

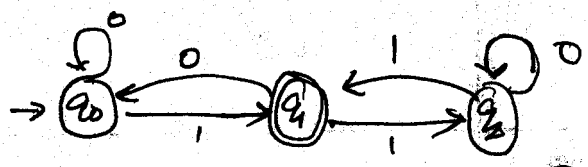
Example 2.6 $L = \{awaaw_2a : w_1, w_2 \in \{a,b\}^*\}$ is regular.

L regular $\Rightarrow$ $L^1, L^2, L^3 \ldots$ are also regular.



EXERCISES

① Which of following are accepted by

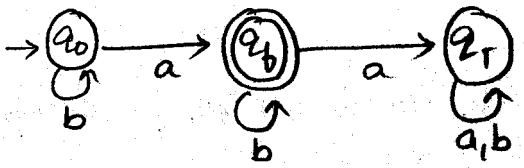


0001 ✓ 0000110. ✗

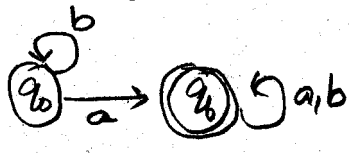
01001 ✓

② For $\Sigma = \{a,b\}$ construct dfa's

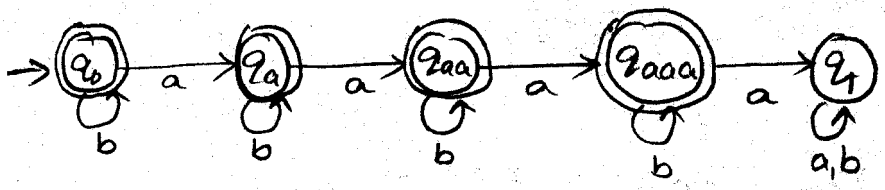
(a) all strings with exactly one a.



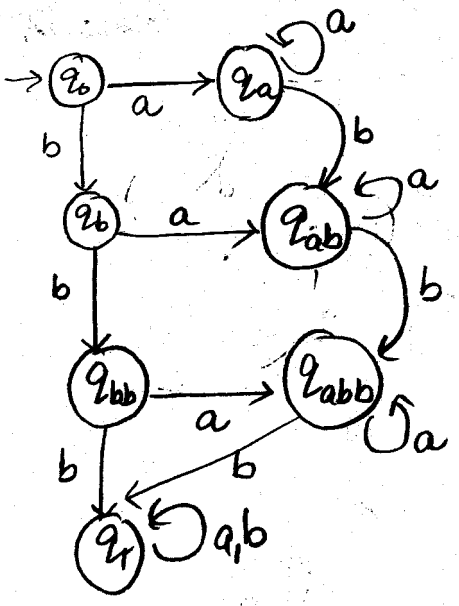
(b) all strings with @least one a



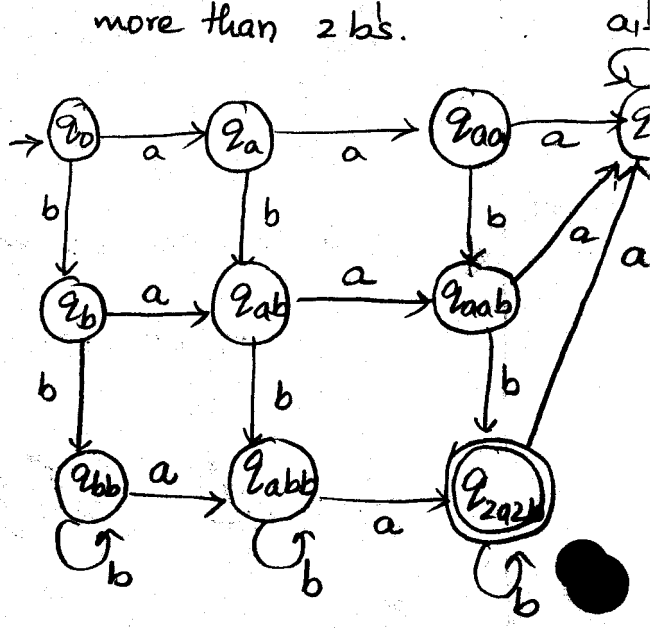
(c) all strings with no more than 3 a's.



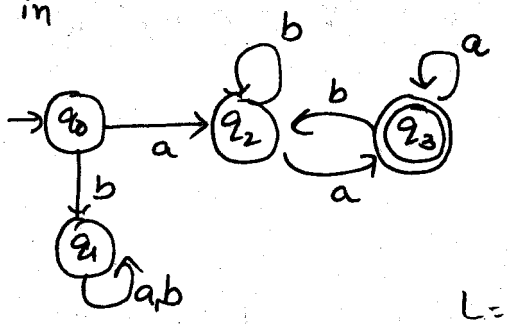
(d) @least one a & exactly 2 b's.



(e) all strings with 2 a's & more than 2 b's.



(3) ST in



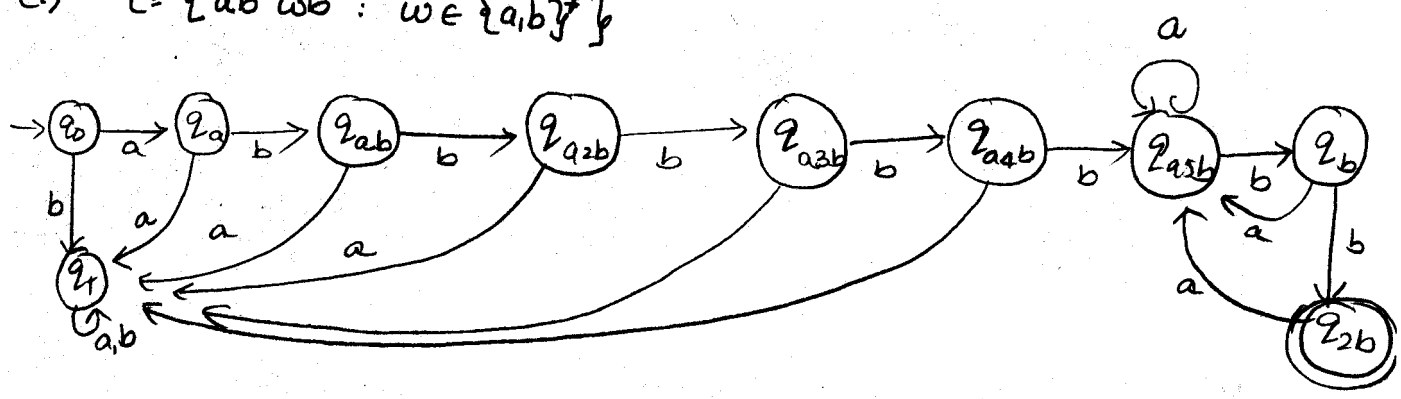
If: $q_3 \notin F$
$q_0, q_1, q_2 \in F$, resulting dfa accepts $\bar{L}$.

$L = \{awa : w \in \Sigma^*\}$

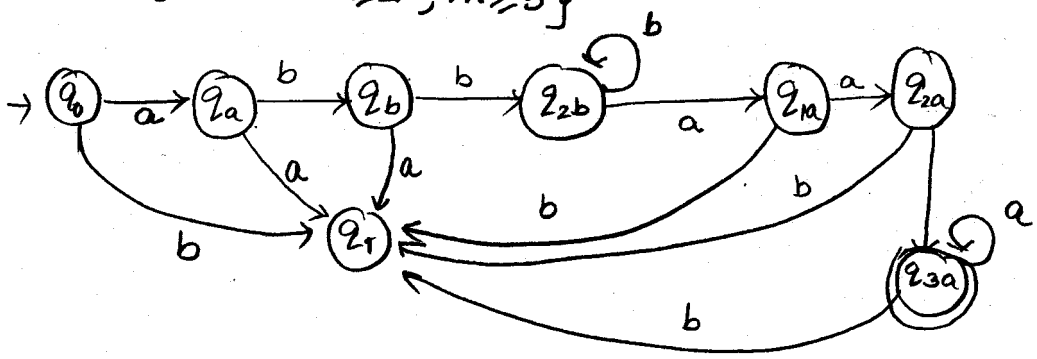
$\bar{L}$: is accepted by the changes to L.

(4) $M = (Q, \Sigma, \delta, q_0, F)$
$\hat{M} = (Q, \Sigma, \delta, q_0, Q-F)$ then $\overline{L(M)} = L(\hat{M})$

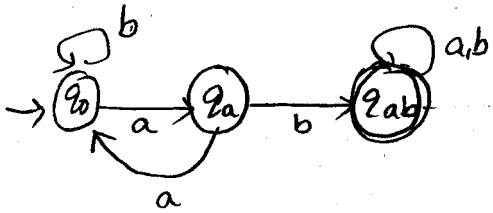
(5) (a) $L = \{ab^5wb^2 : w \in \{a,b\}^*\}$



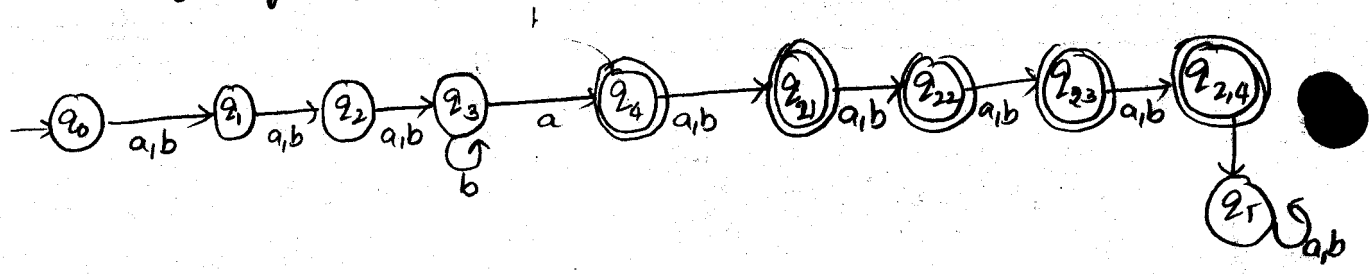
(b) $L = \{ab^na^m : n \geq 2, m \geq 3\}$



(c) $L = \{w_1abw_2 : w_1 \in \{a,b\}^*, w_2 \in \{a,b\}^*\}$

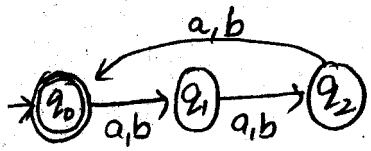


(6) $\Sigma = \{a,b\}$ give dfa for $L = \{w_1 a w_2 : |w_1| \geq 3, |w_2| \leq 5\}$

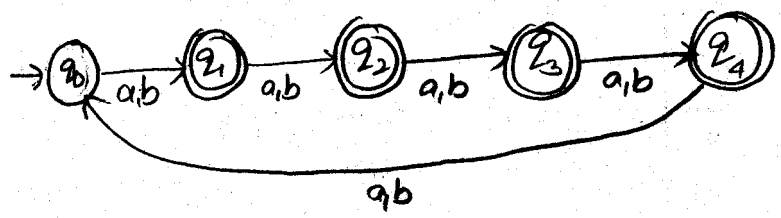


(7) on $\Sigma = \{a,b\}$

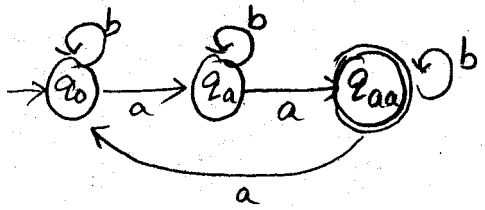
(a) $L = \{w : |w| \bmod 3 = 0\}$



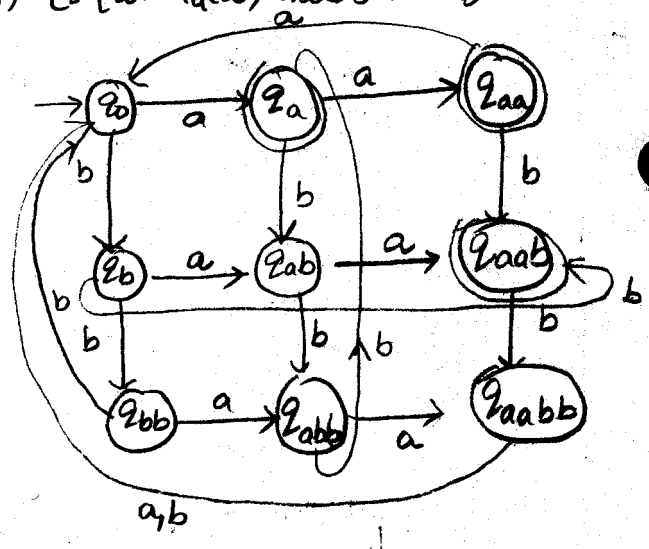
(b) $L = \{w : |w| \bmod 5 \neq 0\}$



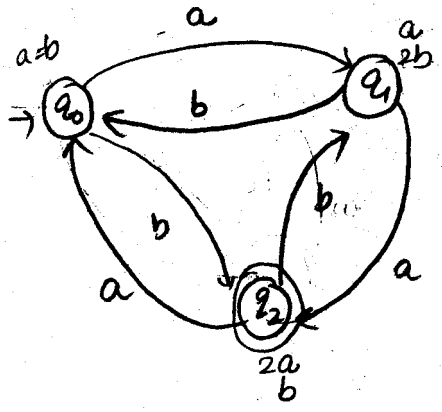
(c) $L = \{w : n_a(w) \bmod 3 > 1\}$



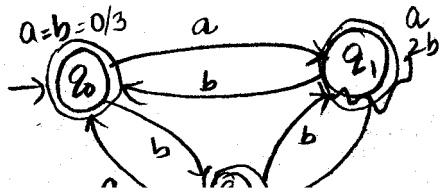
(d) $L = \{w : n_a(w) \bmod 3 > n_b(w) \bmod 3\}$



(e) $L = \{w : (n_a(w) - n_b(w)) \bmod 3 > 0\}$

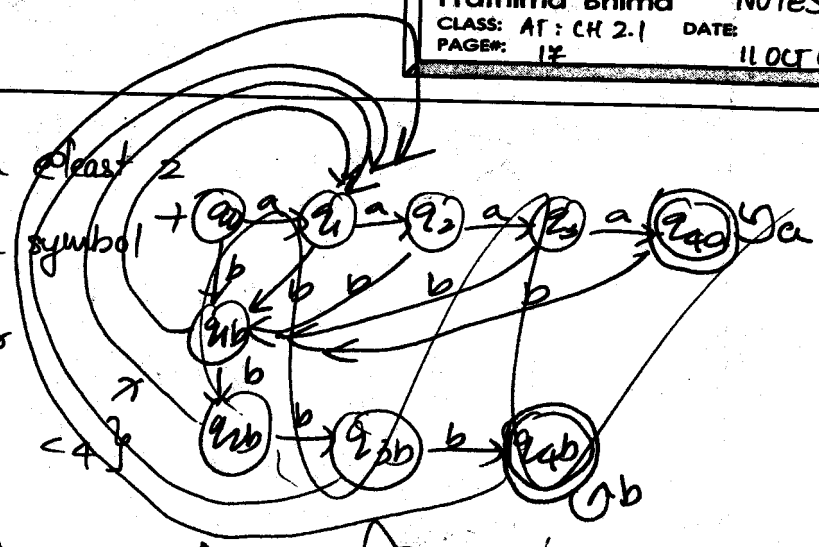


(f) $L = \{w : (n_a(w) + 2n_b(w)) \bmod 3 < 2$.

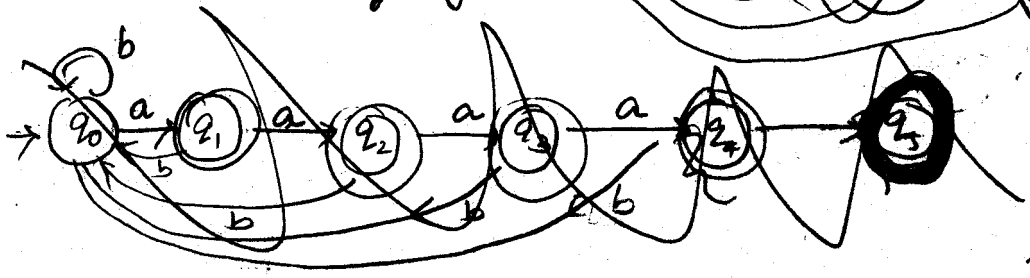


(8) run = substring of length at least 2 & entirely of same symbol

on $\Sigma = \{a,b\}$ find dfa for

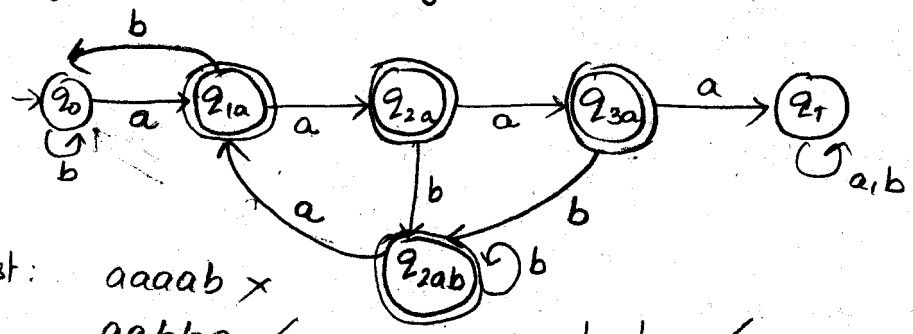


(a) $L = \{w$: no runs of ~~exactly~~ length $< 4\}$



zero, one a should be accepted

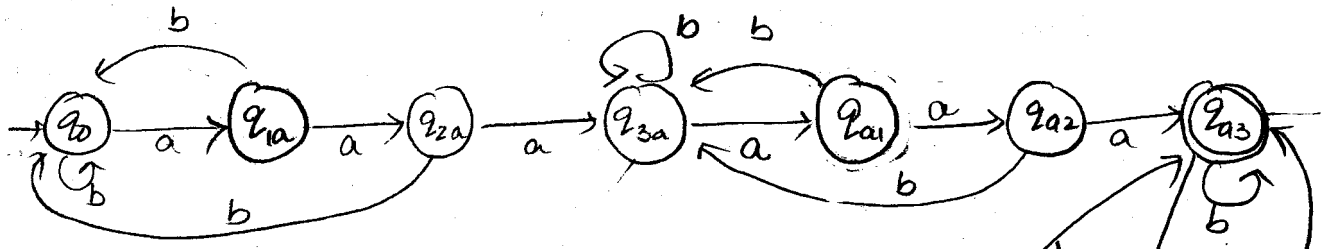
(b) $L = \{w$: every run of a's has length 2 or 3. $\}$ implies one a also accepted



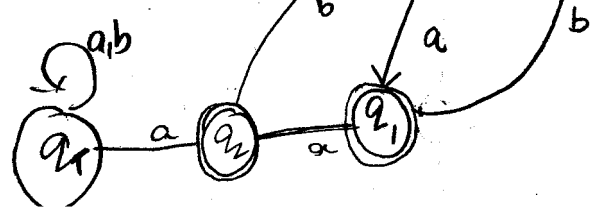
Test: aaaab ✗
aabba ✓
baaab ✓
baba ✓
baaaa ✗

(c) $L = \{w$: ~~[illegible]~~ exactly 2 runs of a's of length 3. $\}$

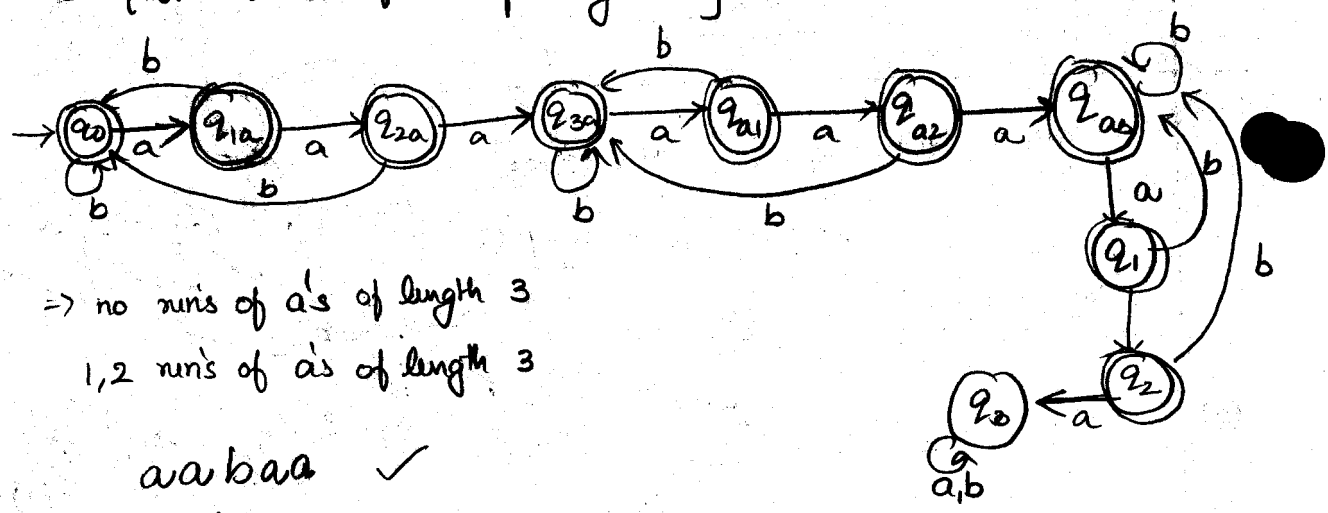
waaaw / waaawaaaw



Test: aaabaaa ✓ aabaaabaaa
aaabaaabaa



(d) $L = \{w :$ @most 2 runs of a's of length 3 $\}$



=> no runs of a's of length 3

1,2 runs of a's of length 3

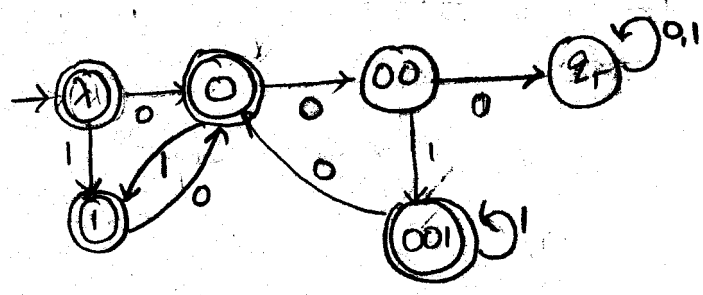
aabaa ✓

aaab

aaabaaa

(9) $\Sigma = \{0,1\}$

(a) Every 00 followed by 1



Test:

00100111 ✓

010001 ✗

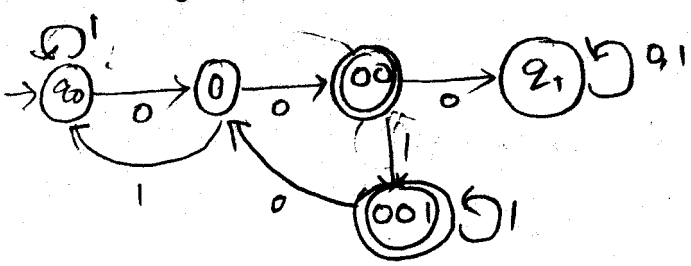
01 ✓

10 ✓

1 ✓

0 ✓

(b) All strings containing 00 but not 000.

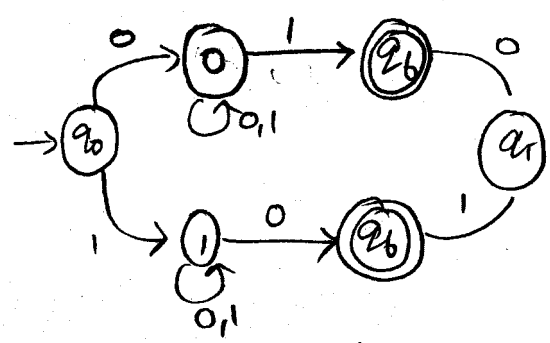


Test: 0010011 ✓

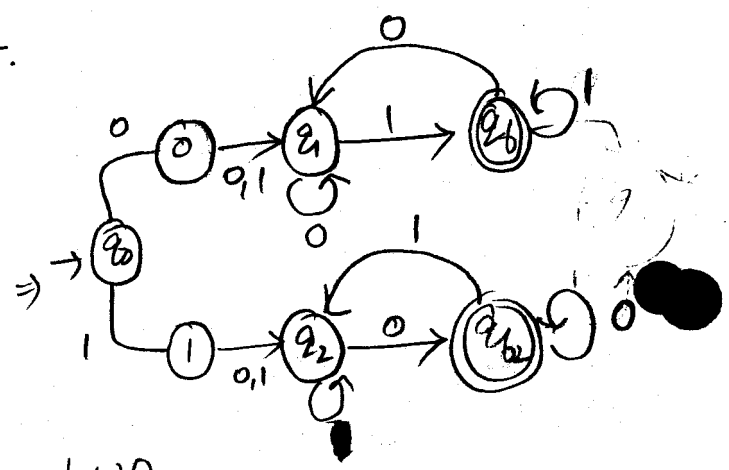
00011 ✓

1001000 ✗

(c) leftmost differs from eightmost.



=>



1 w 0

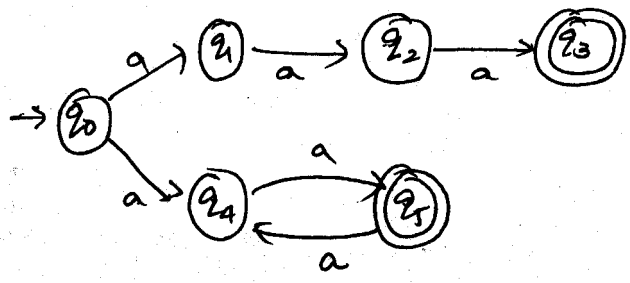
Nfa:

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$\delta : Q \times (\Sigma \cup \{\lambda\}) \to 2^Q$$

Example 2.7

fig 2.8



Example 2.9

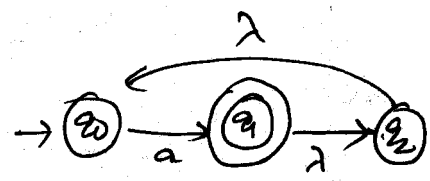
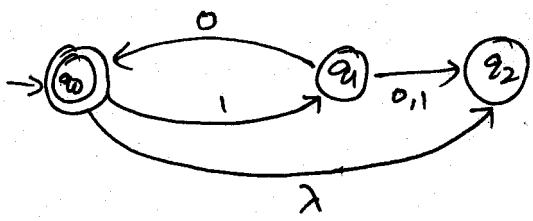


Fig 2.10

Example 2.8

Fig 2.9

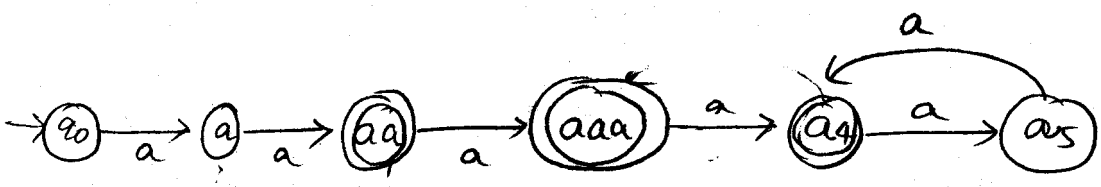


$$L(M) = \{ w \in \Sigma^* : \delta^*(q_0, w) \cap F \neq \emptyset \}$$

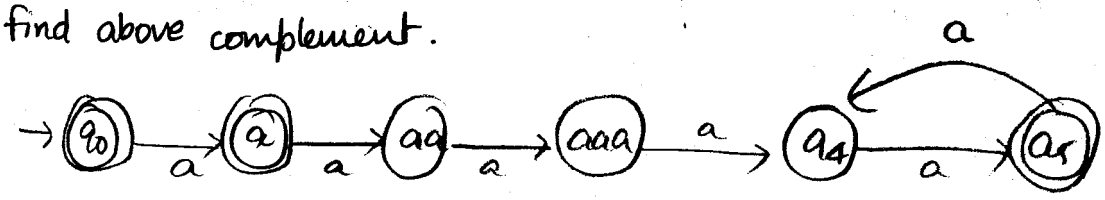
EXERCISES

(2) find dfa defined by: fig 2.8

L: $\{aaa\} \cup \{a^{2n} : n \geq 1\}$



(3) find above complement.



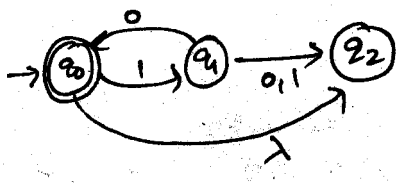
(4) Fig 2.9 $\delta^*(q_0, 1011)$: $\delta^*(q_1, 011) \to \delta^*(q_0, 11) \to q_2$

$\delta^*(q_1, 01)$: q_2

(5) Fig 2.10 : $\delta^*(q_0, a)$ $\{q_0, q_1, q_2\}$

$\delta^*(q_1, \lambda)$ $\{q_0, q_2\}$

(6) for:



$\delta^*(q_0, 1010) \to \delta^*(q_1, 010) \to \delta^*(q_0, 10) \to \delta^*(q_1, 0) \to \{q_0\ q_2\}$ $\to q_2$

$\to \delta^*(q_2, 1010)$ $\times$ $\to \delta^*(q_2, 10)$ $\times$ $\delta^*(q_2, 10)$ $\times$

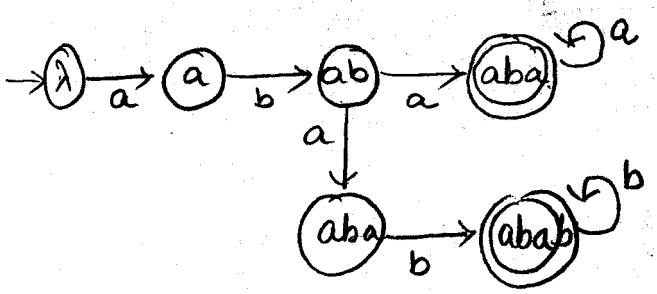
$\delta^*(q_1, 00)$

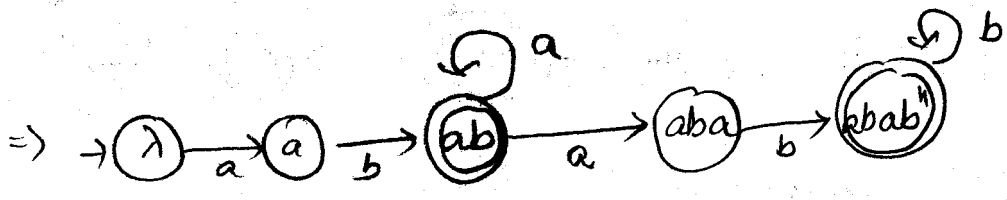
$\delta^*(q_1, 00) \to \delta^*(q_0, 0) \to q_2$

$\delta^*(q_2, 0)$ $\times$

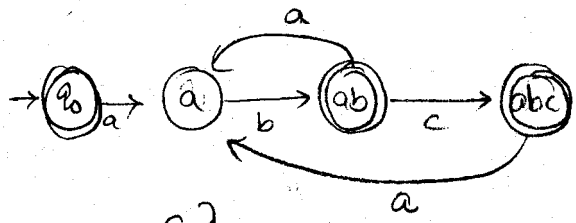
(7) no more than 5 states, design nfa for

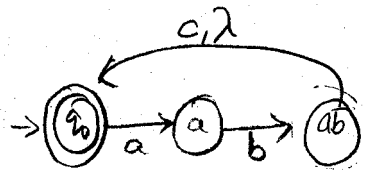
$\{abab^n : n \geq 0\} \cup \{aba^n : n \geq 0\}$





(8) Construct nfa with 3 states for $\{ab, abc\}^*$





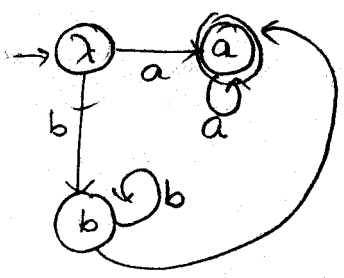
(9) Can-I be done in fewer states? than 3?

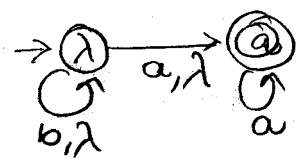
No as $|abab^n|$ least $= 3$ for $n=0$.

(10) (a) find nfa with 3 states that accepts

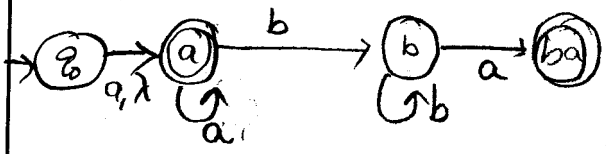
$L = \{a^n : n \geq 1\} \cup \{b^m a^k : m \geq 0, k \geq 0\}$

(b) can fewer than 3 states be possible?

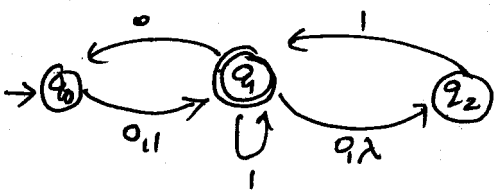




(11) Nfa - 4 states for $L = \{a^n : n \geq 0\} \cup \{b^n a : n \geq 1\}$



(12)



00 : $\delta^*(q_0, 00) \to \{q_0\ q_2\} \cap F = \emptyset$ reject

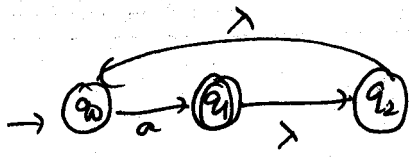
01001 : $\{q_1\} \cap F \neq \emptyset$ accept

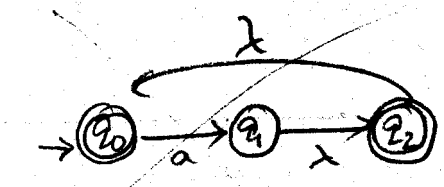
10010 : $\{q_0, q_2\} \cap F = \emptyset$ reject

000 : $\{q_1, q_2\} \cap F \neq 0$ accept

0000 : $\{q_0\ q_2\}$ reject

(13) What is complement of

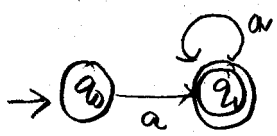




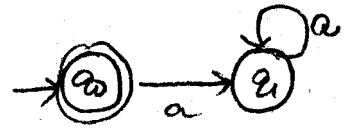
$\{\lambda, a, aa, \ldots\}$

cant take complement of nfa

dfa



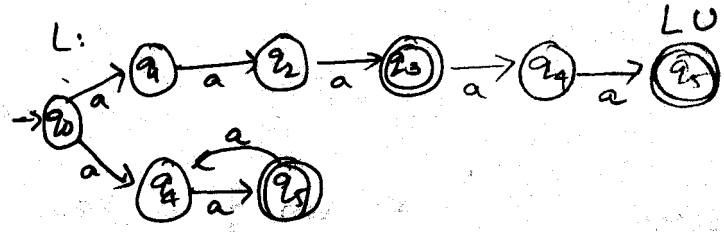
complement →



all strings $n_a(w) > 1$

all strings $n_a(w) < 1$

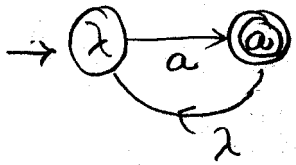
(14) L:



$L \cup \{a^5\}$?

(15) 13 ? $\{a\}^* - \{a\}^+$

(16) Find nfa for $\{a\}^*$ such that removing one edge will accept $\{a\}$



(17) Can above be done with dfa ?

No : need two paths to accept one a / more a's

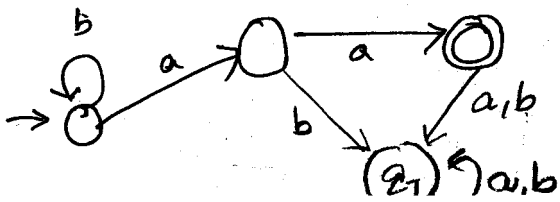
(18) let $p_0 \in Q_0$.

$\delta(p_0, \lambda) \rightarrow Q_0$

if Q_0 not initial, equivalent to M.

(19) Yes.

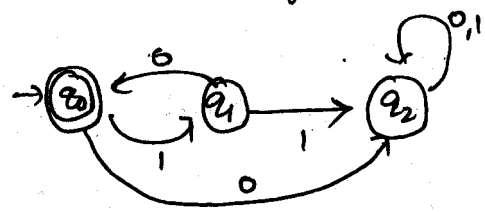
(20)



Equivalence of Nfa & Dfa:

fig 2.11

dfa

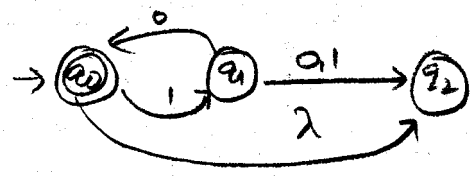


$\{\lambda, 10, 1010 \ldots\}$

$\Rightarrow \{(10)^n : n \geq 0\}$

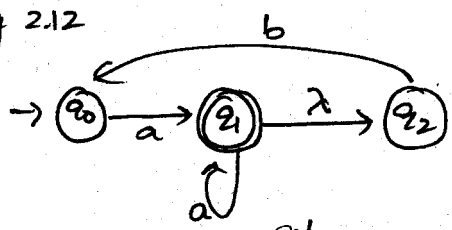
$\simeq$

nfa



$\{(10)^n : n \geq 0\}$

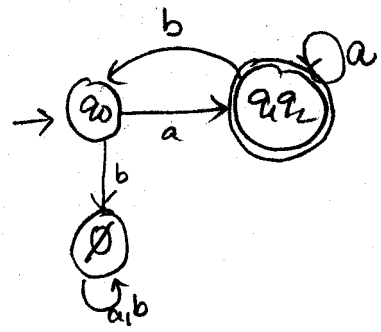
fig 2.12



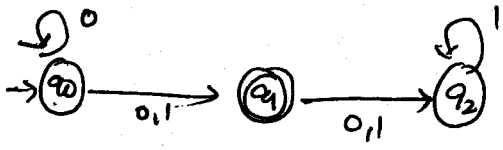
nfa

	a	b
$\{q_0\}$	$\{q_1, q_2\}$	$\emptyset$
F $\{q_1 q_2\}$	$\{q_1, q_2\}$	$\{q_0\}$
$\emptyset$	$\emptyset$	$\emptyset$

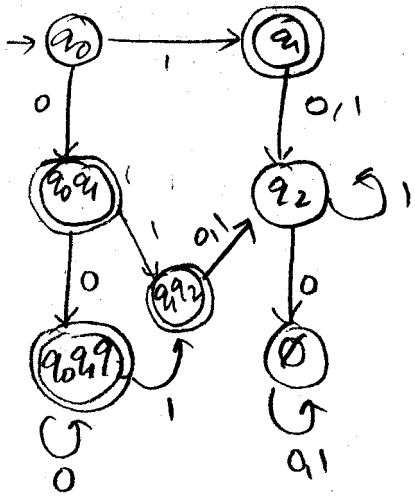
dfa =



Example 2.13

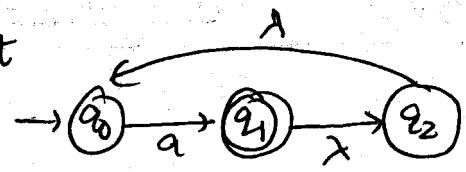


	0	1
$\{q_0\}$	$\{q_0 q_1\}$	$\{q_1\}$
$\{q_0 q_1\}$	$\{q_0 q_1 q_2\}$	$\{q_1 q_2\}$
$\{q_1\}$	$\{q_2\}$	$\{q_2\}$
$\{q_2\}$	$\emptyset$	$\{q_2\}$
$\{q_1 q_2\}$	$\{q_2\}$	$\{q_2\}$
$\{q_0 q_1 q_2\}$	[illegible]	[illegible]

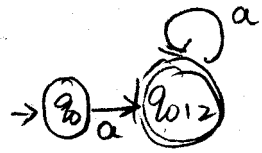


If too many states getting combined, don't end, go on & enumerate all

2-3 EXERCISES

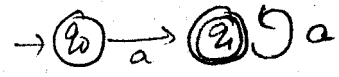
① Convert  to dfa. Is there a simpler way?

	a	
$\{q_0\}$	$\{q_0 q_1 q_2\}$	
$\{q_0 q_1 q_2\}$	$\{q_0 q_1 q_2\}$	

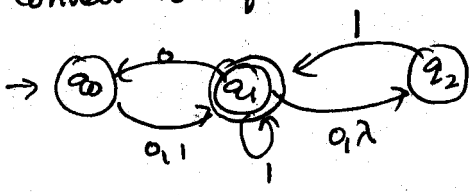


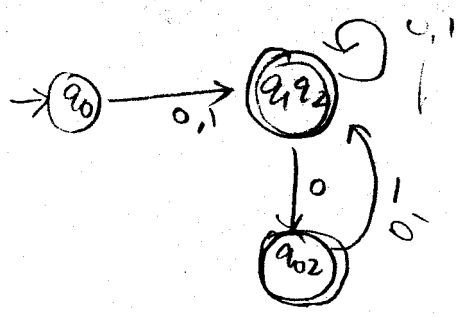
Yes: directly:

as $L = \{a^n : n > 0\}$



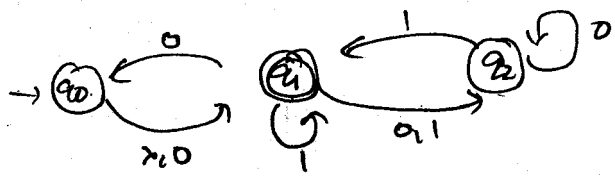
② Convert to dfa





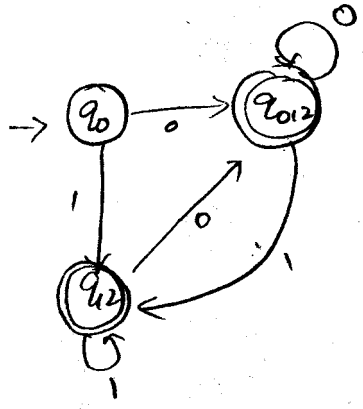
	0	1
$\{q_0\}$	$\{q_1 q_2\}$	$\{q_1 q_2\}$
$\{q_1 q_2\}$	$\{q_0 q_2\}$	$\{q_1 q_2\}$
$\{q_0 q_2\}$	$\{q_1 q_2\}$	$\{q_1 q_2\}$

③ Convert nfa → dfa



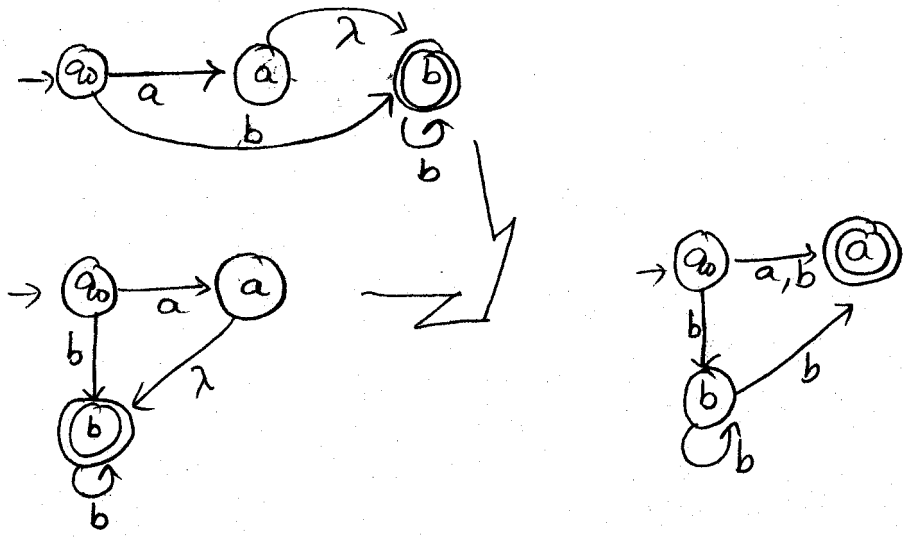
$q_0 = q_1$

	0	1
$\{q_0\}$	$\{q_0 q_1 q_2\}$	$\{q_1 q_2\}$
$\{q_1 q_2\}$	$\{q_0 q_1 q_2\}$	$\{q_1 q_2\}$
$\{q_0 q_1 q_2\}$	$\{q_0 q_1 q_2\}$	$\{q_1 q_2\}$



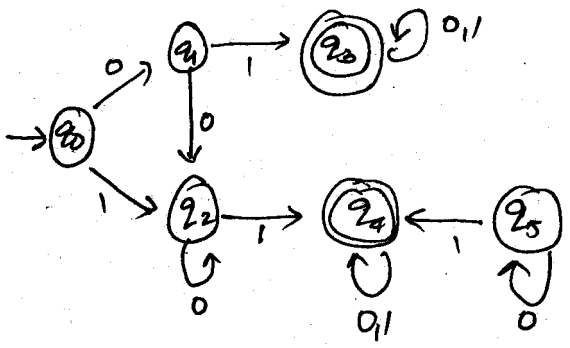
(8) find nfa without λ-Transitions, single final state for

$\{a\} \cup \{b^n : n \geq 1\}$



$\boxed{\text{CH \# 2.4}}$ (Reduction of states in dfa)

Example 2.17



	q_0	q_1	q_2	q_3	q_4
q_0		D	D	D	D
q_1			ID	D	D
q_2				D	D
q_3					ID
q_4					

I states: q_5

D: $\{q_0, q_1\}$
$\{q_0, q_2\}$

ID: $\{q_3\ q_4\}$
$\{q_1\ q_2\}$

$\{q_0\}, \{q_1 q_2\}, \{q_3\ q_4\}$

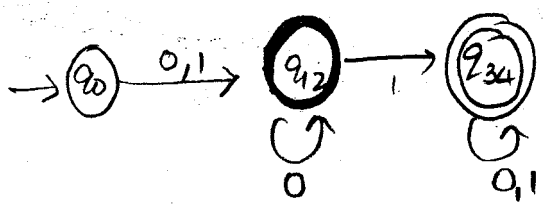
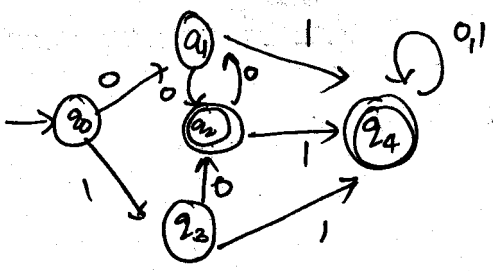


fig: 2.18

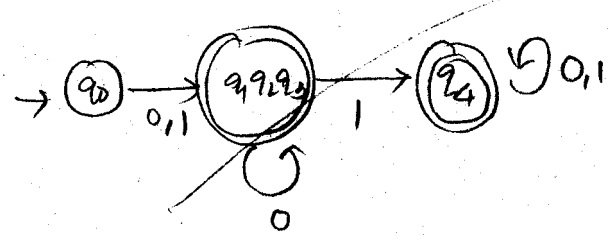


D: $\{q_1\ q_4\}$
$\{q_2, q_4\}$
$\{q_3, q_4\}$
$\{q_0, q_4\}$

ID: $\{q_1\ q_3\}$
$\downarrow 0 \quad \downarrow$
$\{q_2\}\ \{q_4\}$

	q_0	q_1	q_2	q_3	q_4
q_0		D	D	D	D
q_1			ID	ID	D
q_2				ID	D
q_3					D
q_4					

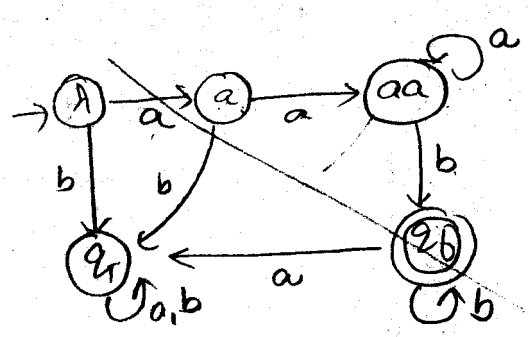
$\{q_0\}, \{q_1 q_2 q_3\}\ \{q_4\}$



EXERCISES

(2) (a) find minimal dfa for

$L = \{a^n b^m : n \geq 2, m \geq 1\}$

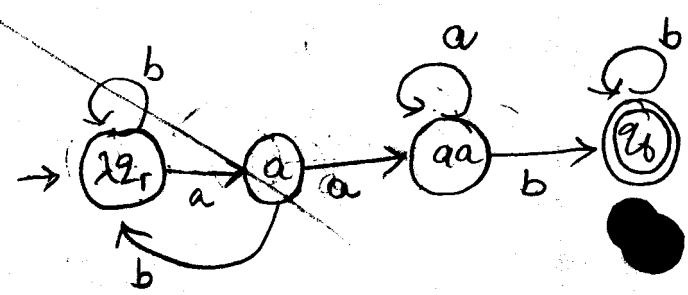


	λ	a	aa	q_f	q_T
λ		D	D	D	ID
a			D	D	D
aa				ID	D
q_f					D
q_T					

$\{\lambda, q_T\}, \{a\}, \{a\}, \{aa\}\ \{q_f\}$

ID: (aa, q_f)
(λ, q_T)

D: (a, q_f) (λ, q_f) $(q_T\ aa)$
(q_T, q_f) (λ, aa) (q_f, aa)

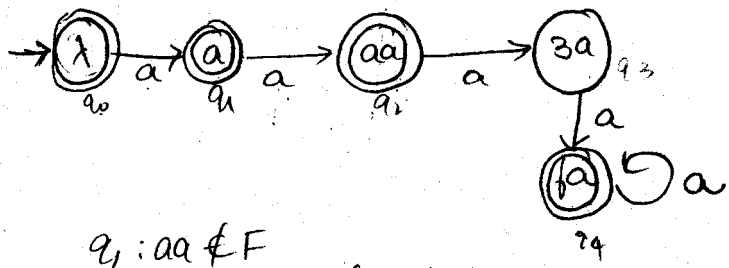


② (a) $L = \{a^n b^m : n \geq 2, m \geq 1\}$

minimal

(b) $L = \{a^n b : n \geq 0\} \cup \{b^n a : n \geq 1\}$

(c) $L = \{a^n : n \geq 0, n \neq 3\}$



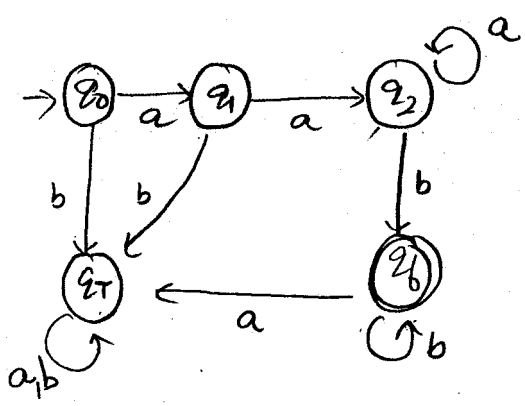
	λ	a	aa	3a	fa
λ		D	D	D	D
a			D	D	D
aa				D	D
3a					D
fa					

$q_1 : aa \notin F$
$(q_4, aa) \in F$: $(q_1 q_4)$ D

$(\lambda, aaa) \notin F$: $(\lambda, 3a)$,
$(3a, aaa) \in F$ D

already in minimal //

(a)



	q_0	q_1	q_2	q_f	q_T
q_0		D	D	D	D
q_1			D	D	D
q_2				D	D
q_f					D
q_T					

$q_f \in F$
rest $\notin F$: D-states

∴ minimal

CHAPTER : 3

RL & RG

→ $L(r_1+r_2) = L(r_1) \cup L(r_2)$

$L(r_1 . r_2) = L(r_1) . L(r_2)$

$L(r_1^*) = (L(r_1))^*$

Example 3.2

$L(a^* . (a+b))$?

$L(a^* . (a+b)) = L(a^*) . L(a+b)$

$= L(a^*) . \{L(a) \cup L(b)\}$

$= L(a^*) . \{a, b\}$

$= \{\lambda, a, aa \ldots\} . \{a, b\}$

$= \{a\}^+ \cup \{a^n b^{n+1} : n \geq 0\}$

Example 3.3

$\Sigma = \{a, b\}$

$r = (a+b)^* . (a+bb)$

$\{a, b\}^* . \{a, bb\}$: $= \{a, bb, aa, abb, ba, bbb \ldots\}$

Example 3.4

$r = (aa)^* (bb)^* b$

$L = \{(aa)^n (bb)^n b : n \geq 0\}$

$L = \{a^{2n} b^{2m+1} : n, m \geq 0\}$

Example 3.5

$\Sigma = \{0, 1\}$: w has atleast one pair of consecutive zeroes.

$(0+1)^* 00 (0+1)^*$

3.6. no consecutive zeroes : $(1+01)^* (0+\lambda)$

EXERCISES

① $L\left[(a+b)^* b (a+ab)^*\right]$ Find strings $|w| < 4$.

$\{\lambda, a, b, ab, ba, aa, bb, aba, baa, aaa, bba, bab, abb, aab, \ldots\}$. b.

$\{\lambda, a, ab, aab, aba, aaa\}$ - - -

$|w| < 4$: $\{b, ab, bb, ba, bab \ldots\}$

② $((0+1)(0+1)^*)^* \, 00 \, (0+1)^*$ denote @least one pair of consecutive 0's.

yes.

③ $r = (1+01)^* (0+1^*)$ also denotes no consecutive zeroes.

$\downarrow$

$(1+01)^* (0+\lambda+\{1\}^+)$

$((1+01)^* (0+\lambda)) + (1+01)^* \{1\}^+$

$\downarrow$

$(1+01)^*$

$= (1+01)^* (0+\lambda)$ $\Rightarrow$ no consecutive zero's

④ RE? $\{a^n b^m : n \geq 3, m \text{ is even}\}$

$aaa(a^*)(bb)^*$

⑤ RE = ? $\{a^n b^m : (n+m) \text{ is even}\}$

$\left[(aa)^*(bb)^* + (aa)^* a (bb)^* b\right]$

RE = ?

⑥ (a)

$L_1 = \{a^n b^m : n \geq 4, m \leq 3\}$

$aaaa \cdot a^* (\lambda + b + bb + bbb)$

(b) $L_2 = \{a^n b^m : n < 4; m \leq 3\}$

$(\lambda + a + aa + aaa)(\lambda + b + bb + bbb)$

(c) $\bar{L}_1 : \{a^n b^m : n < 4, m > 3\}$ (d)

$\times$ $(\lambda + a + aa + aaa)\, bbbb\, b^* +$

either $n < 4$ or $m \geq 4$ (or) $a^n b^k a^l$

$(\lambda + a + aa + aaa)\, b^* + a^* bbbbb^* +$

$(a+b)^* ba (a+b)^*$

RE($\bar{L}$) = all possible rule breakers in RE(L) →

(d) $\bar{L}_2 : \{a^n b^m : n < 4, m \leq 3\}$

$n \geq 4 \,/\, m > 3$

$\bar{L}_2 : aaaa a^* b^* + a^* bbbb b^* + (a+b)^* ba (a+b)^*$

⑧ $L\left[(aa)^* b (aa)^* + a(aa)^* ba(aa)^*\right]$

wbw : $w : a^{2n} : n \geq 0$

$w : a^{2n+1}$ $n \geq 0$

b having even a's on both ends or

b having odd a's on both ends.

3-1
Exercises :

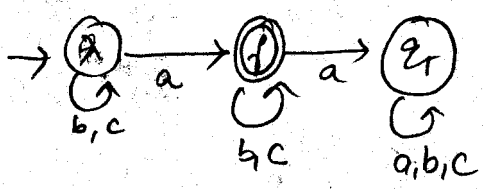
16

(a) $\Sigma = \{a,b,c\}$

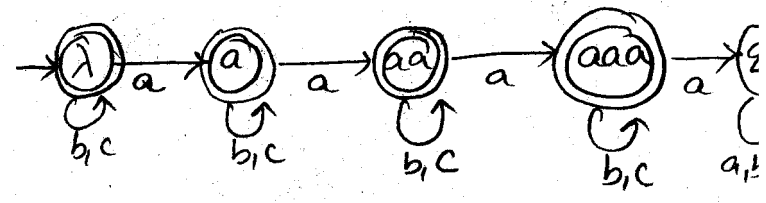
Exactly one a.

$(b+c)^* a (b+c)^*$

a ✓ bcbca ✓
ab ✓ bcaa ✗



(b) no more than 3 a's.



$[(b+c)^* a (b+c)^* a (b+c)^* a (b+c)^*$
$+ (b+c)^* + (b+c)^* a (b+c)^* +$
$(b+c)^* a (b+c)^* a (b+c)^*]$

Test: aa ✓ aaaa ✗
bca ✓

(c) @least one occurance of each symbol in Σ

$(a+b+c)^* a (a+b+c)^* b (a+b+c)^* c (a+b+c)^*$

(d) no run of a's $|w|_a > 2$

$(\lambda + a + aa + b + c)^*$

0, 1, 2

~~$(b+c)^* (\lambda+a)(b+c)^* (\lambda+a)(b+c)^*$~~

(e) run's of a's are multiples of 3.

~~$(b+c)^* aaaa^* (b+c)^*$~~ $aaaa^* (b+c)^*$

$(\lambda + aaaa^* + b + c)^*$

17

(b) Not Ending in 01:

$1^*(0+01+11)^* 11^*(0+\lambda)$ ✓

17.
(a) Ending in 01

$(0+1)^* 01$

(c) Even no. of zeroes

$[1^* 0 1^* 0 1^* + 1]^*$

CHAPTER 3-2

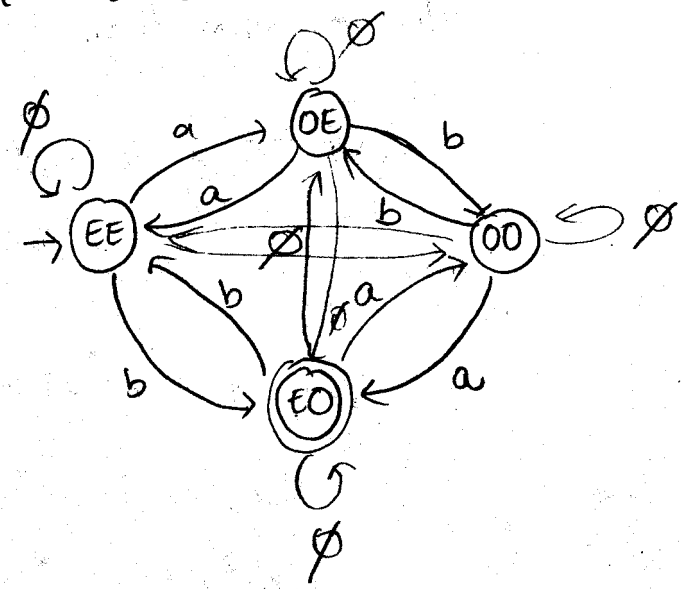
RE = ?

Example 3.11

$L = \{ w \in \{a,b\}^* : n_a(w) \text{ is even}, \; n_b(w) \text{ is odd} \}$

FIND RE for a nfa?

$\boxed{\emptyset \text{ to all unknown moves!}}$



$r + \emptyset = r$

$r\emptyset = \emptyset$

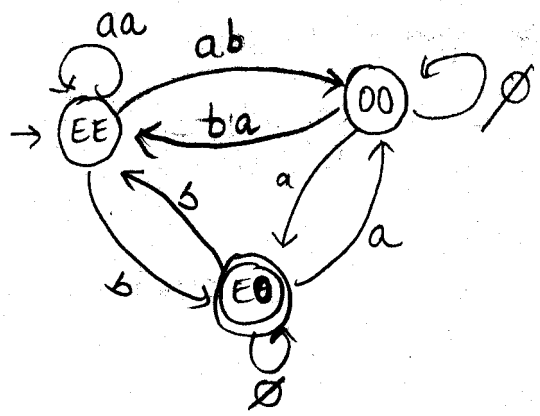
$\emptyset^* = \lambda$

Step ①

$r_{EE} = \emptyset + a\emptyset^* a$

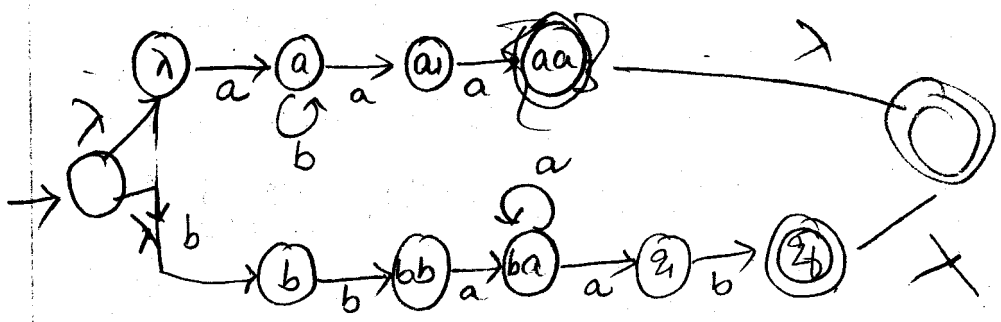
$= aa$

Step 2 $\boxed{\text{now 3-state Rule}}$



EXERCISE

$L(ab^*aa + bba^*ab)$:



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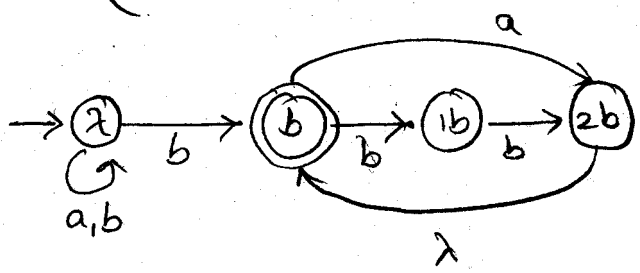
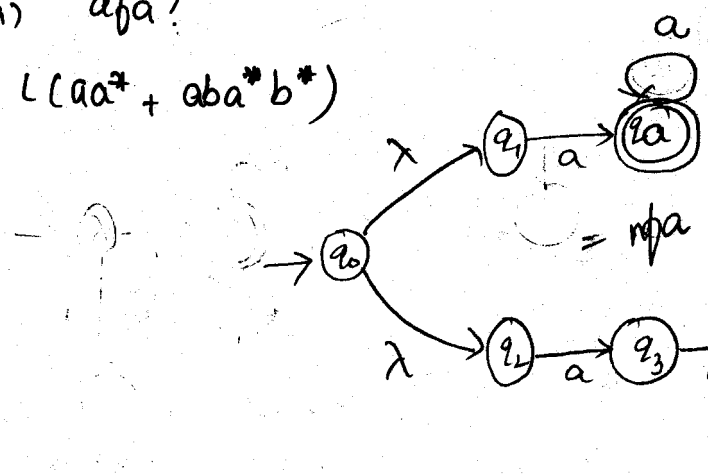
(3) nfa? $((a+b)^* b (a+bb)^*)$

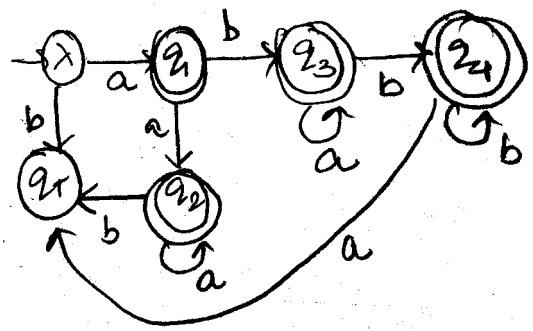


(4) (a) dfa?

$L(aa^* + aba^*b^*)$

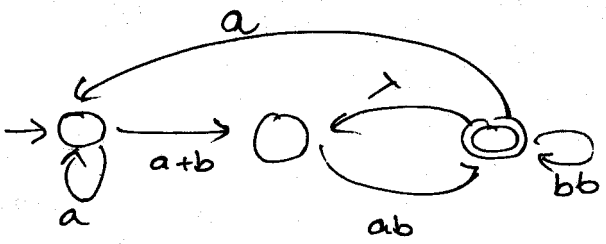


dfa:

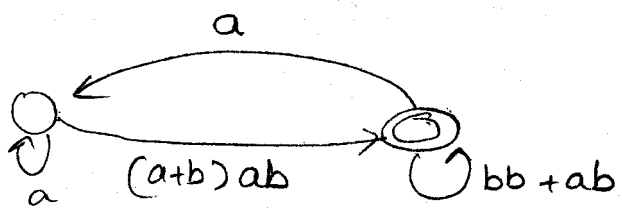


removing an edge in a transition graph ($q_0 \notin F$)

(8)

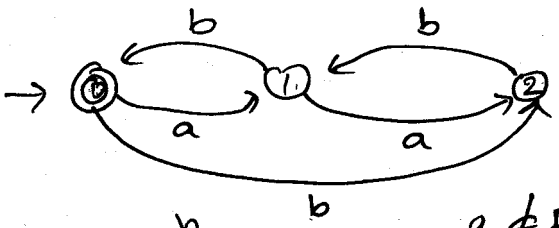


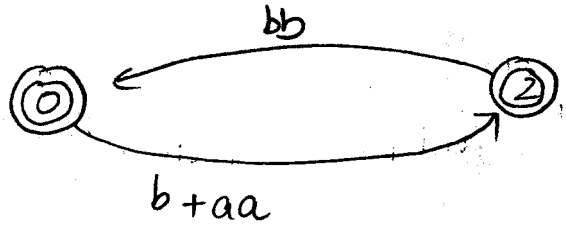
$r = a^* (a+b)ab(ab + bb + aa^*(a+b)ab)^*$



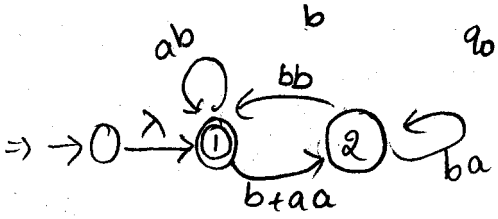
aabbbbb

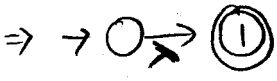
(10) (b)





$q_0 \notin F$.

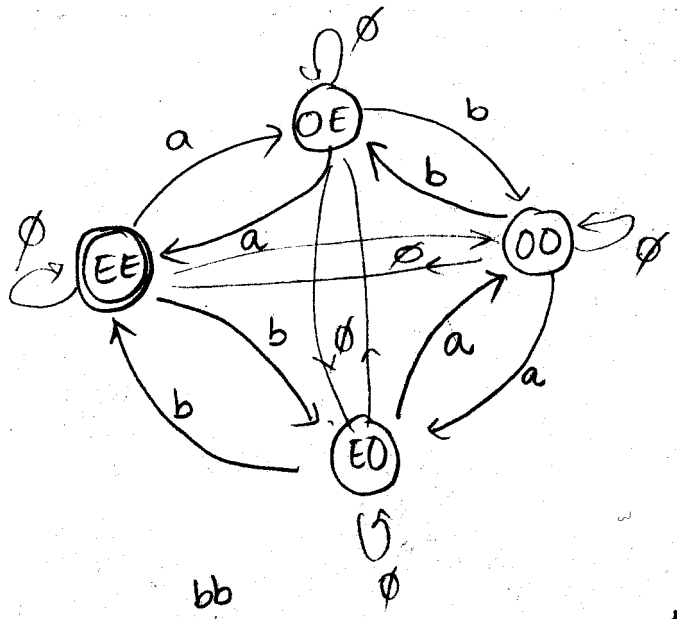


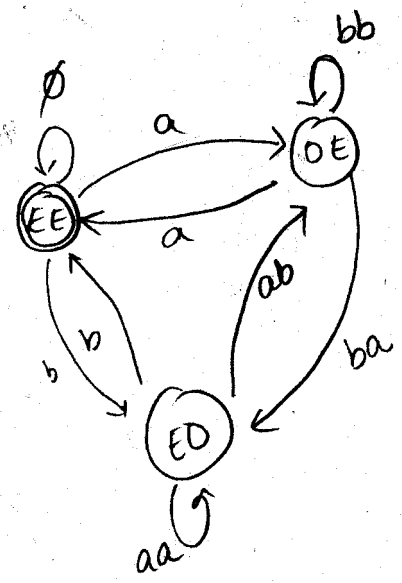


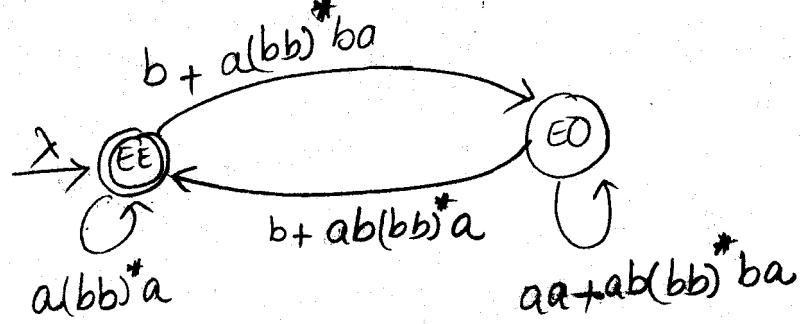
$ab + (b+aa)(ba)^* bb$

RE = ? on $\Sigma = \{a, b\}$

(13) (a) $L = \{w : n_a(w),\ n_b(w)$ are even$\}$







RE :

$$\lambda + [a(bb)^*a][b + a(bb)^*ba][aa + ab(bb)^*ba]^*[b + ab(bb)^*a]$$

λ ✓

aa

CH # 3.3

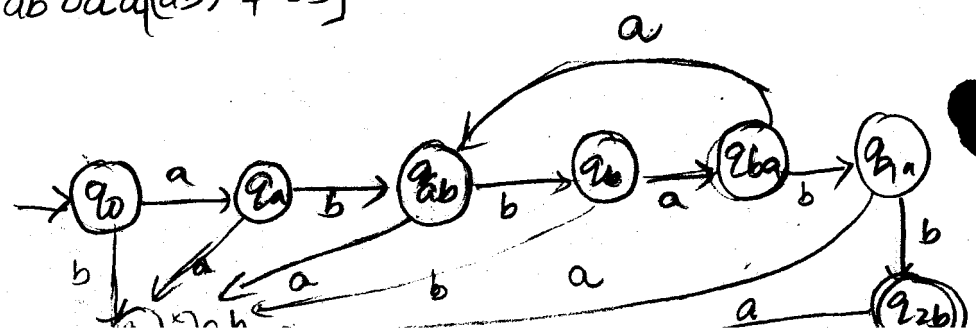
(1) dfa = ?

$S \to abA$

$A \to baB$

$B \to aA / bb$

$ab\,baa[(ab)^* + bb]$



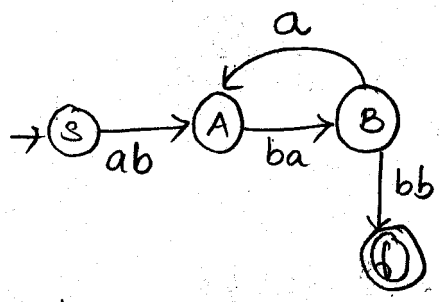
(3) UG = ? for :

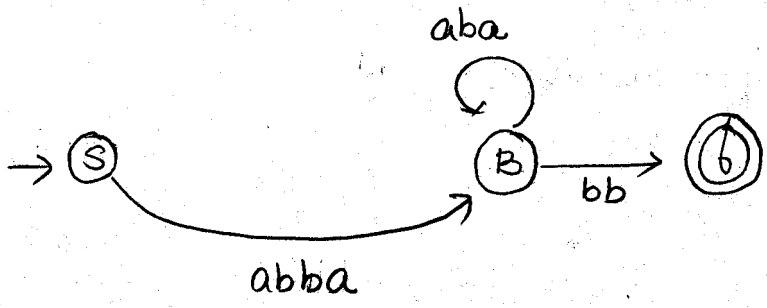
$S \to abA$

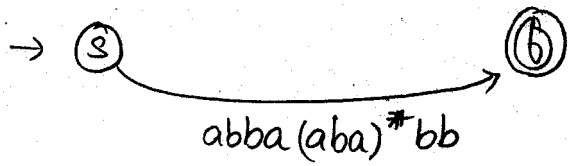
$A \to baB$

$B \to aA / bb$

RE =







Test: abbabb ✓

abbaabaababb ✓

$$\boxed{\begin{array}{l} S \to ABbb \\ A \to abba \\ B \to Baba \;/\; \lambda \end{array}}$$

$S \to ABbb \to abbaBbb \to$

$abba\,Baba\,bb \to$

$abba(aba)^* bb$ ✓

(4) RLG, LLG = ?

$\{a^n b^m : n \geq 2, m \geq 3\}$

RLG:

$S \to aaAB$

$A \to aA / \lambda$

$B \to bbbC$

$C \to bC / \lambda$

LG

$S \to aaA\,bbbB$

$A \to aA / \lambda$

$B \to bB / \lambda$

LLG:

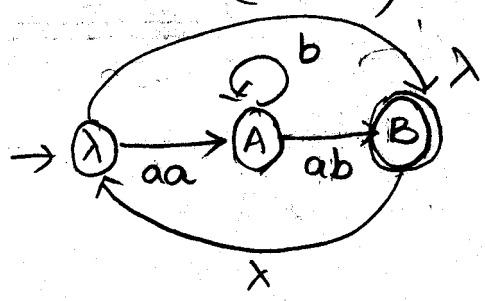
$S \to ABbbb$

$A \to Caa$

$C \to Ca / \lambda$

$B \to Bb / \lambda$

(6) RLG = ? $(aab^*ab)^*$

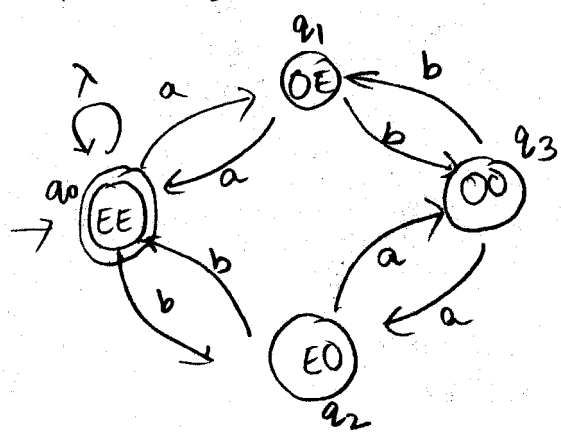


$S \to aaA / \lambda$

$A \to bA / abS$

(13) (a) ~~RLG~~ RG = ? $\Sigma = \{a,b\}$

(a) $n_a(w)$, $n_b(w)$ are even.



$q_0 \to aq_1 / \lambda / bq_2$

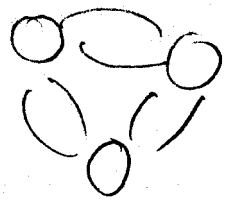
$q_1 \to bq_3 / aq_0$

$q_2 \to aq_3 / bq_0$

$q_3 \to aq_2 / bq_1$

$(n_a - n_b) \bmod 3 = 1$

draw



Right Quotient : L_1/L_2

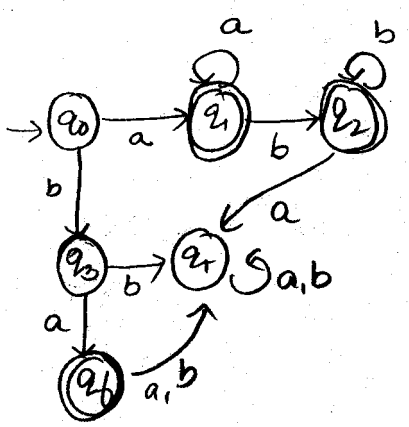
L_1: $\{a^n b^m : n \geq 1, m \geq 0\} \cup \{ba\}$

L_2: $\{b^m : m \geq 1\}$

RIGHT QUOTIENT L_1/L_2

→ draw dfa for L_1

→ ∀ nodes apply L_2 to check fin state



for L_1/L_2

final states are:

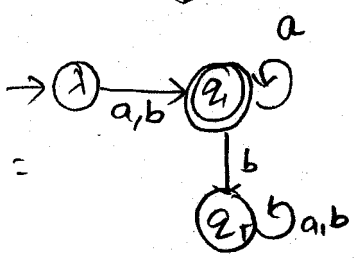
q_1, q_2

Exercises

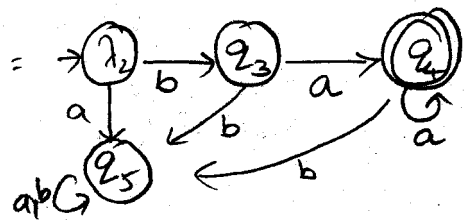
(2) (a) $(a+b)a^* \cap (baa^*)$

$n(a) = ?$

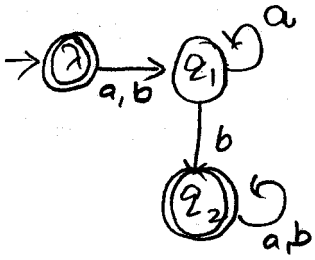
$L_1 =$



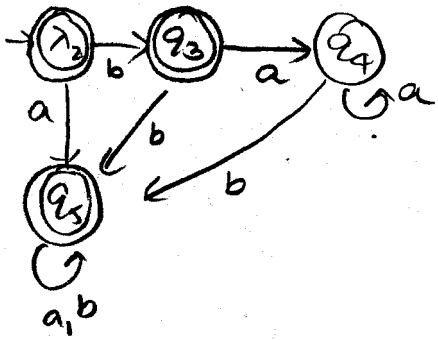
$L_2 =$



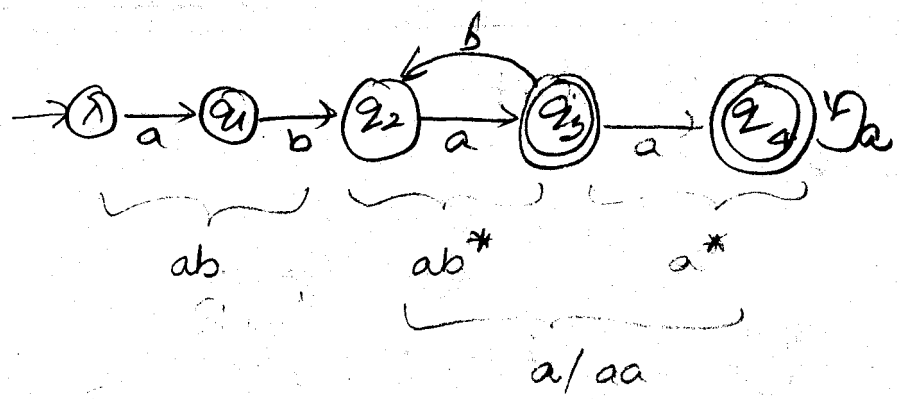
$\overline{L_1} =$



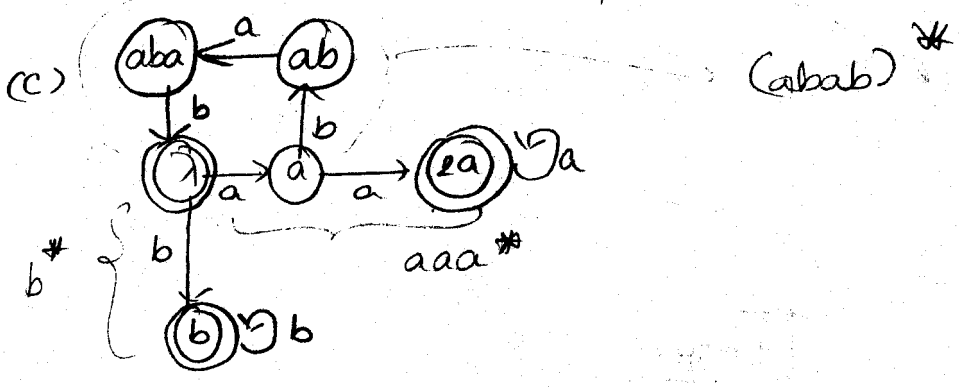
$\overline{L_2} =$



④ (b) $ab(a+ab)^*(a+aa)$

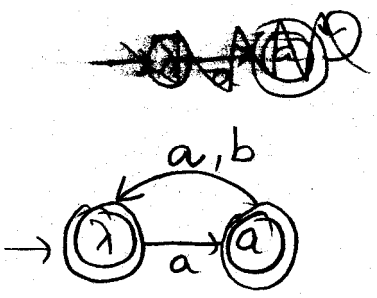


(c)



$(abab)^*$

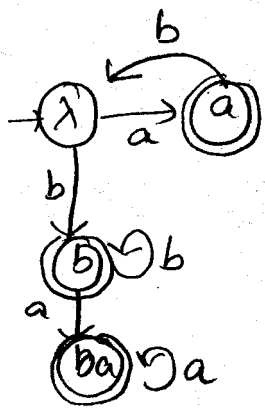
(d)



λ, a, aa, ab, aba

$(aa^*b$

⑤ (a)



ab^*a^*

a ✓

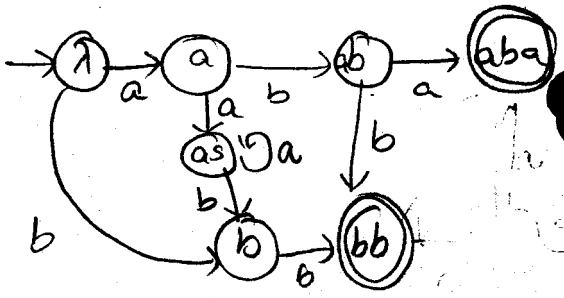
aba ✓

abba ✓

ab^*a

abb

(b) ∅



⑦ bb ✓

abb ✓

a^*bb

ab^*ba

aba ✓

abba ✓

abbba

a^*bba